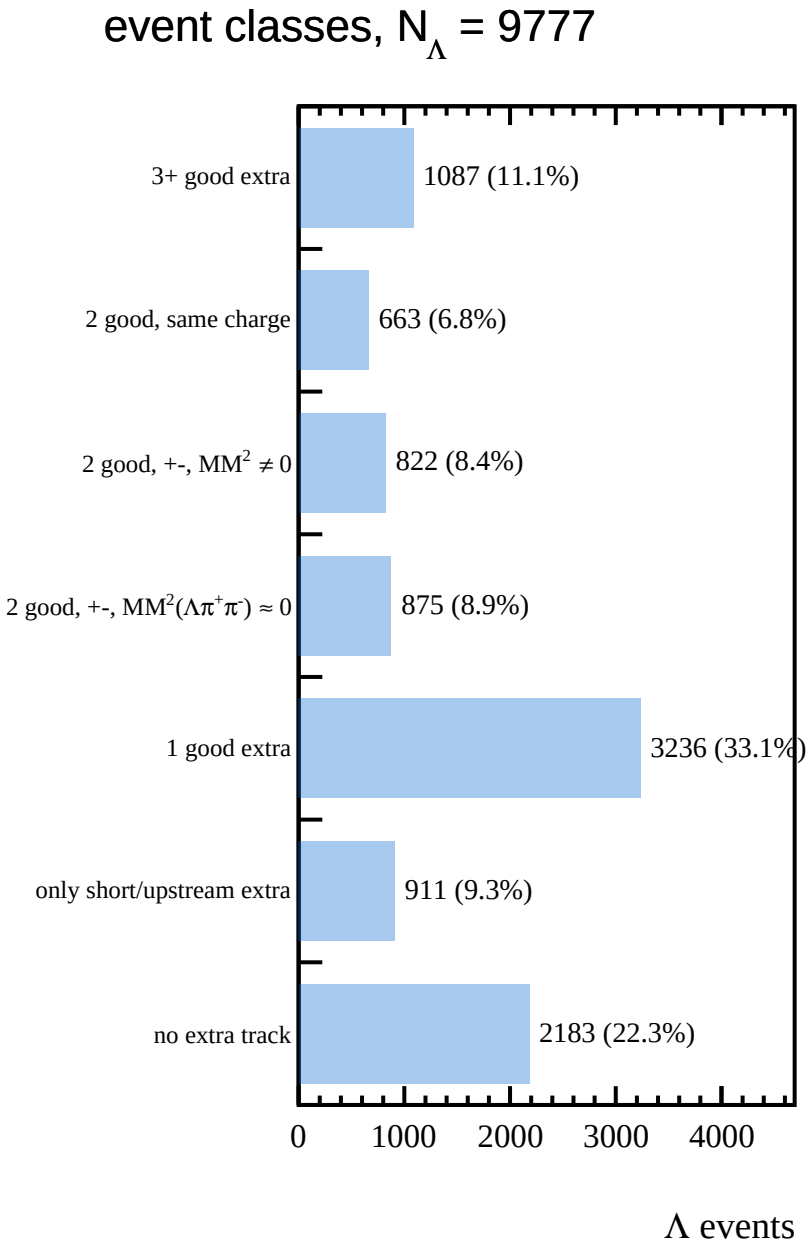
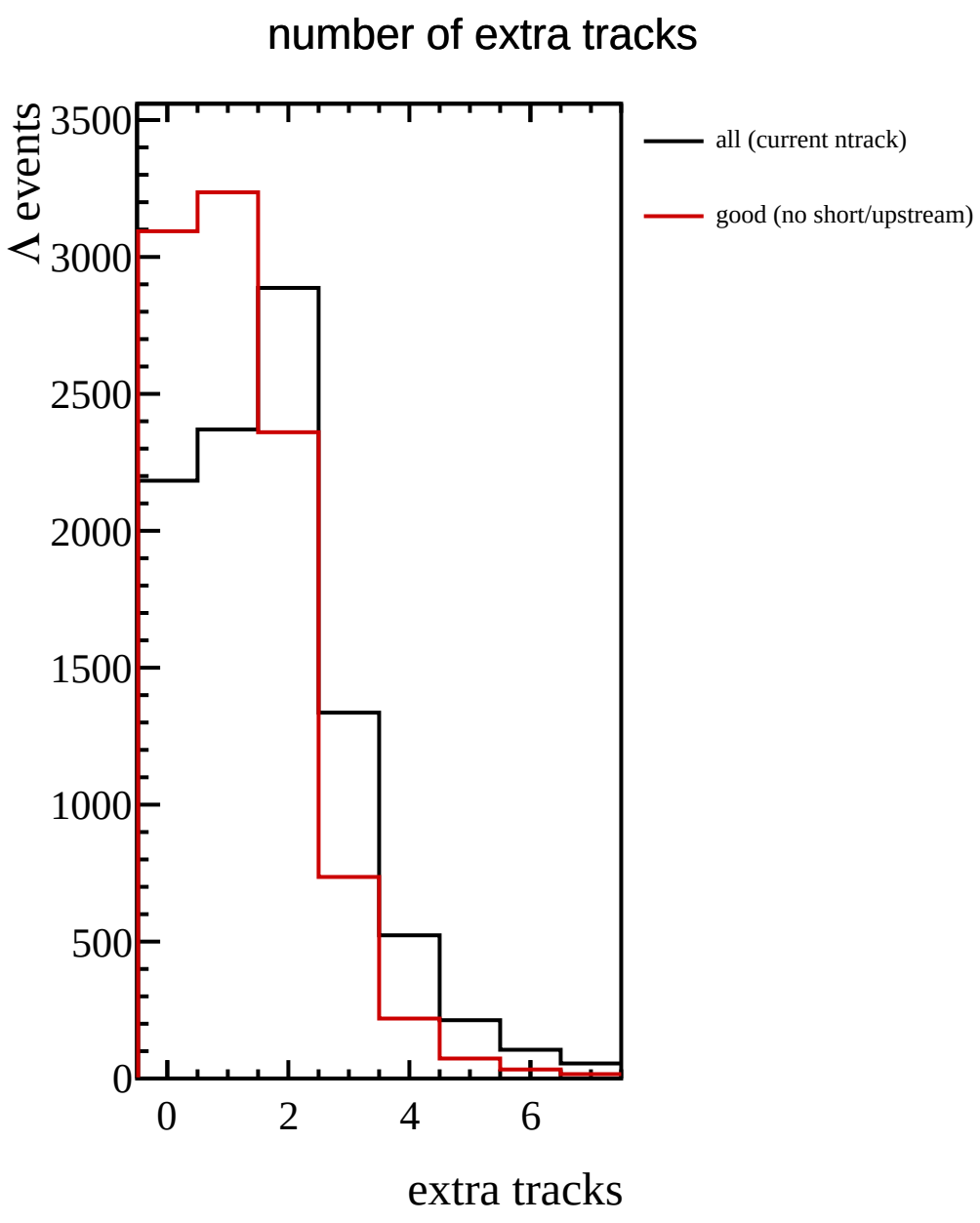
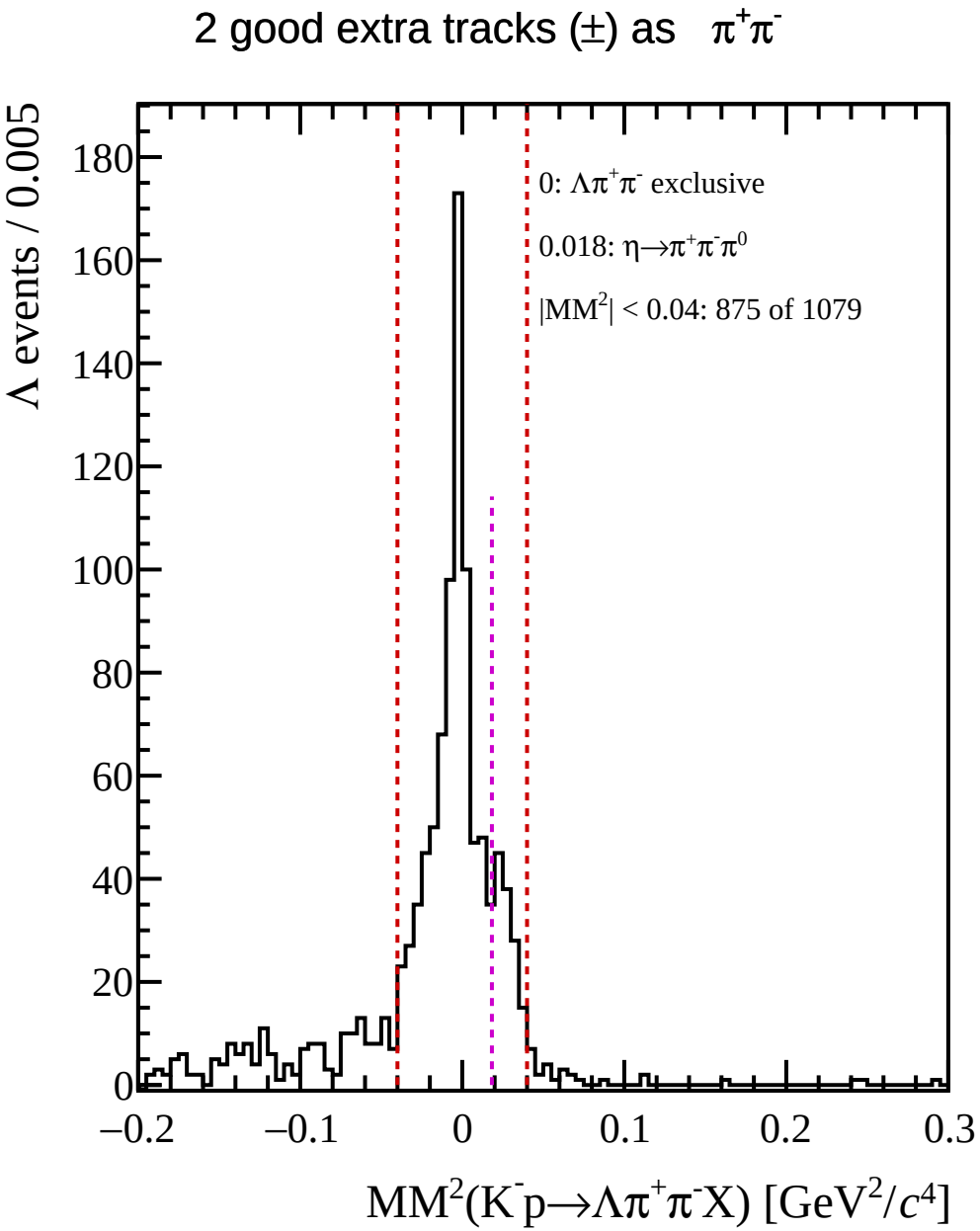
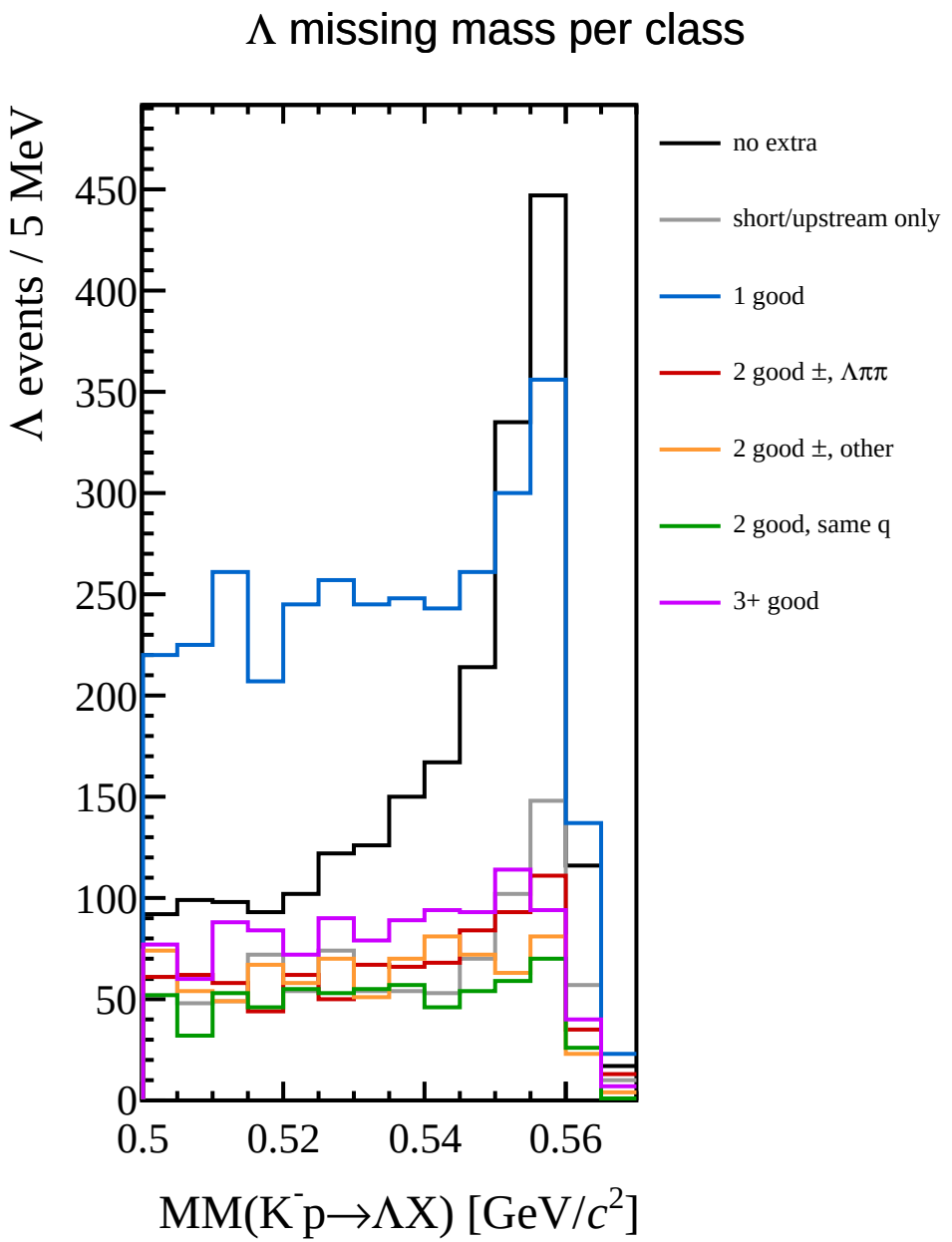
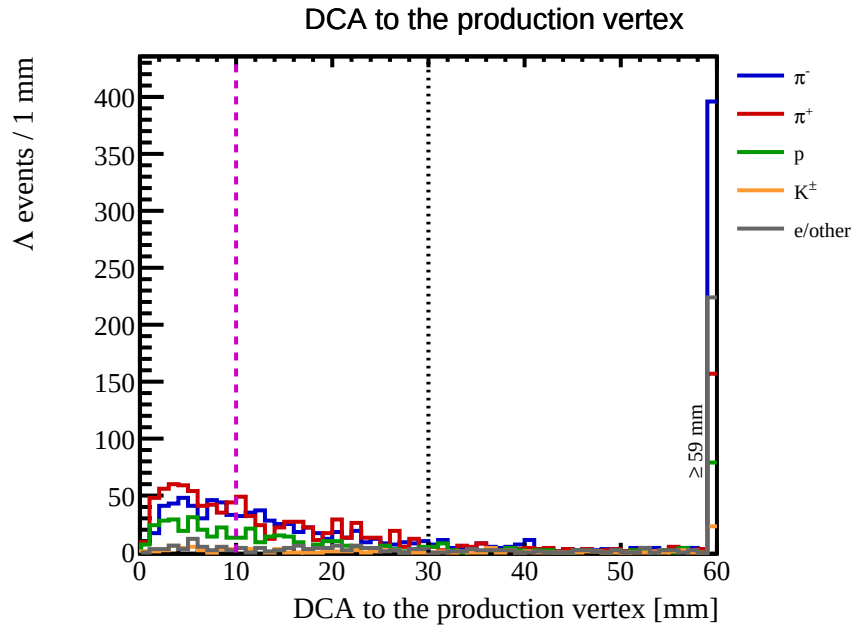

event classes, N_Λ = 9777
3+ good extra
1087 (11.1%)
2 good, same charge
663 (6.8%)
2 good, +-, MM² ≠ 0
822 (8.4%)
2 good, +-, MM²(Λπ⁺π⁻) ≈ 0
875 (8.9%)
1 good extra
3236 (33.1%)
only short/upstream extra
911 (9.3%)
no extra track
2183 (22.3%)
0
1000
2000
3000
4000
Λ events

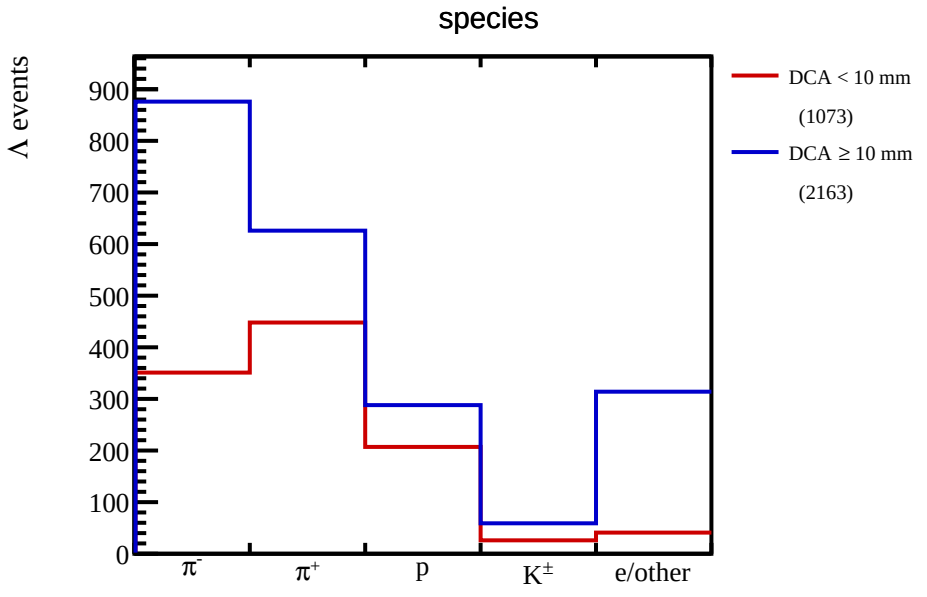

number of extra tracks
Λ events
3500
3000
2500
2000
1500
1000
500
0
0
2
4
6
extra tracks
all (current ntrack)
good (no short/upstream)

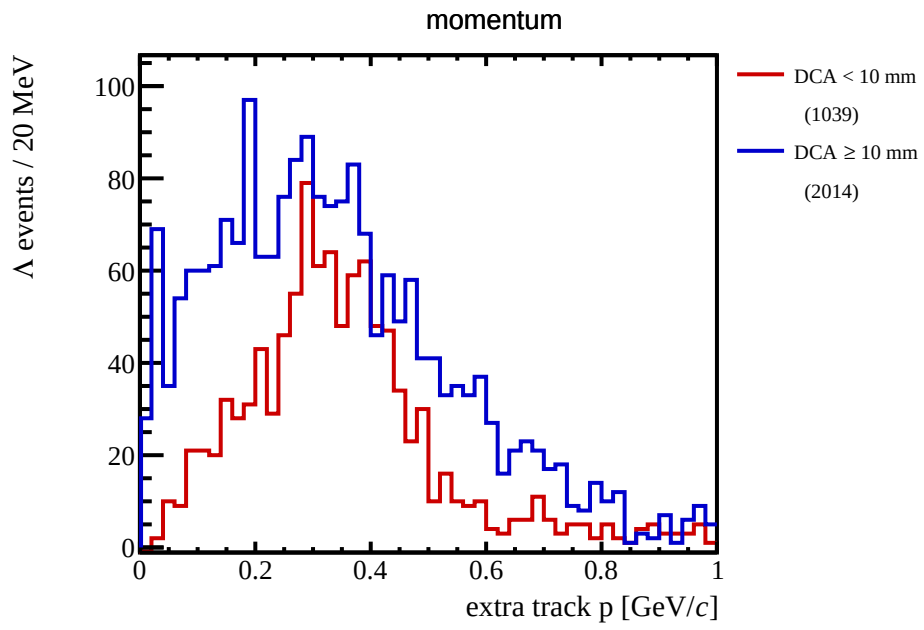

2 good extra tracks (±) as π+π-
Λ events / 0.005
0: Λπ+π- exclusive
0.018: η→π+π-π0
|MM2| < 0.04: 875 of 1079
MM2(K-p→Λπ+π-X) [GeV2/c4]

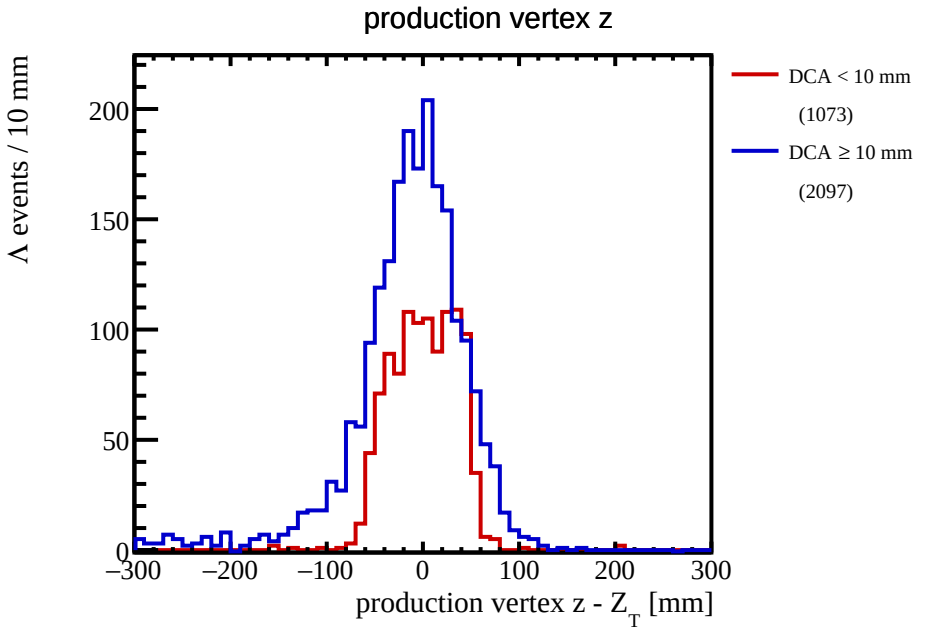

Λ missing mass per class
Λ events / 5 MeV
no extra
short/upstream only
1 good
2 good ±, Λππ
2 good ±, other
2 good, same q
3+ good
MM(K-p→ΛX) [GeV/c2]

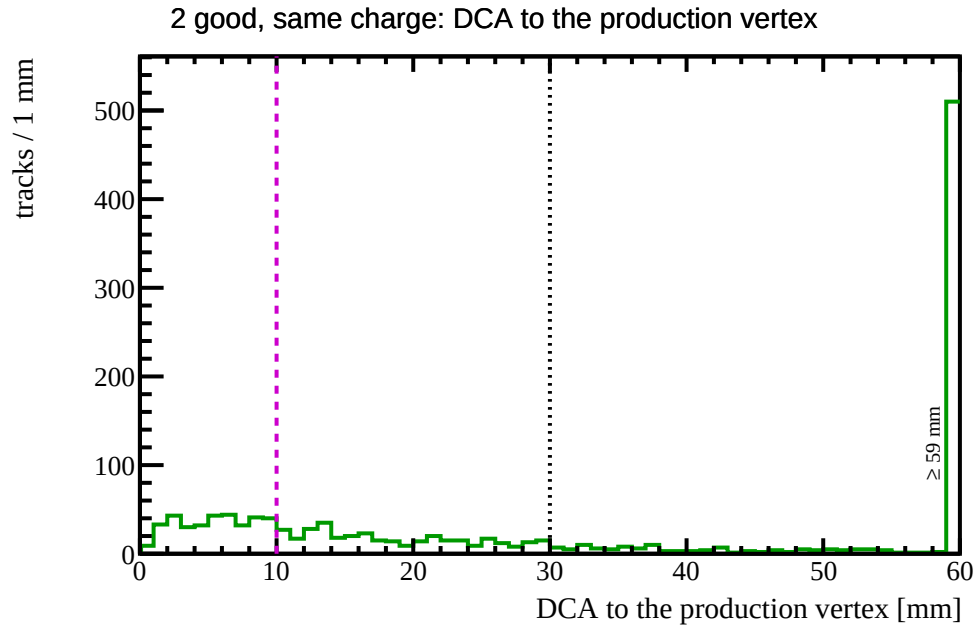
run02447-02452-n5: DCA, χ^2, MM ≥ 0.5, Λ 1.105-1.125 minus sidebands | 1 good extra track

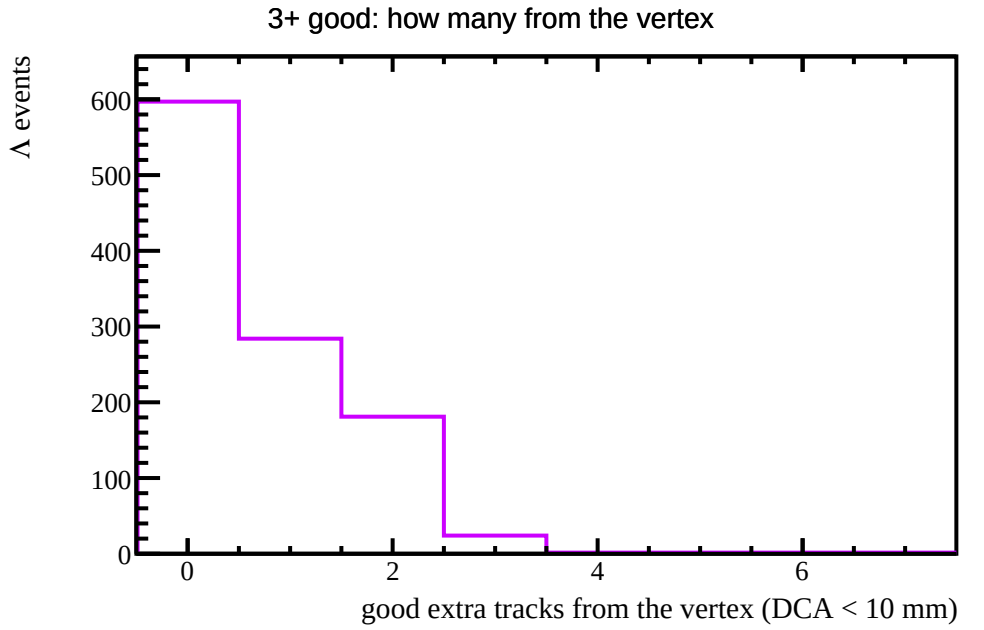


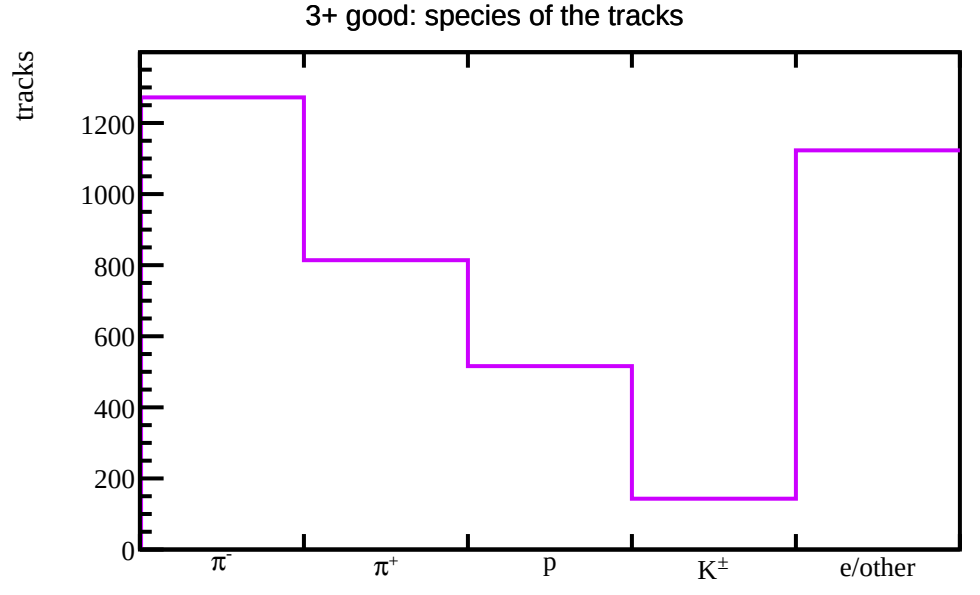


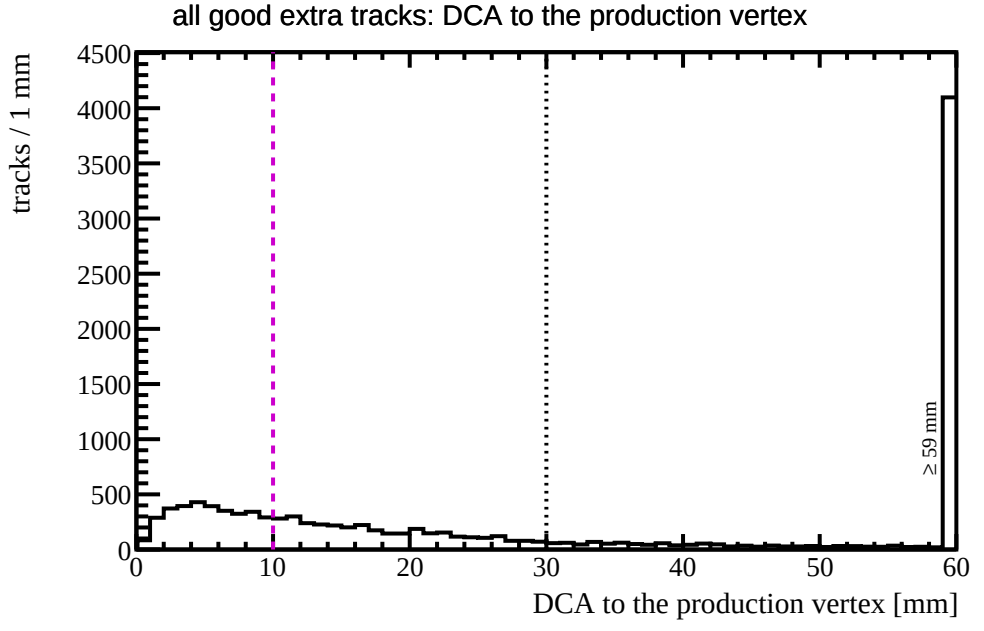


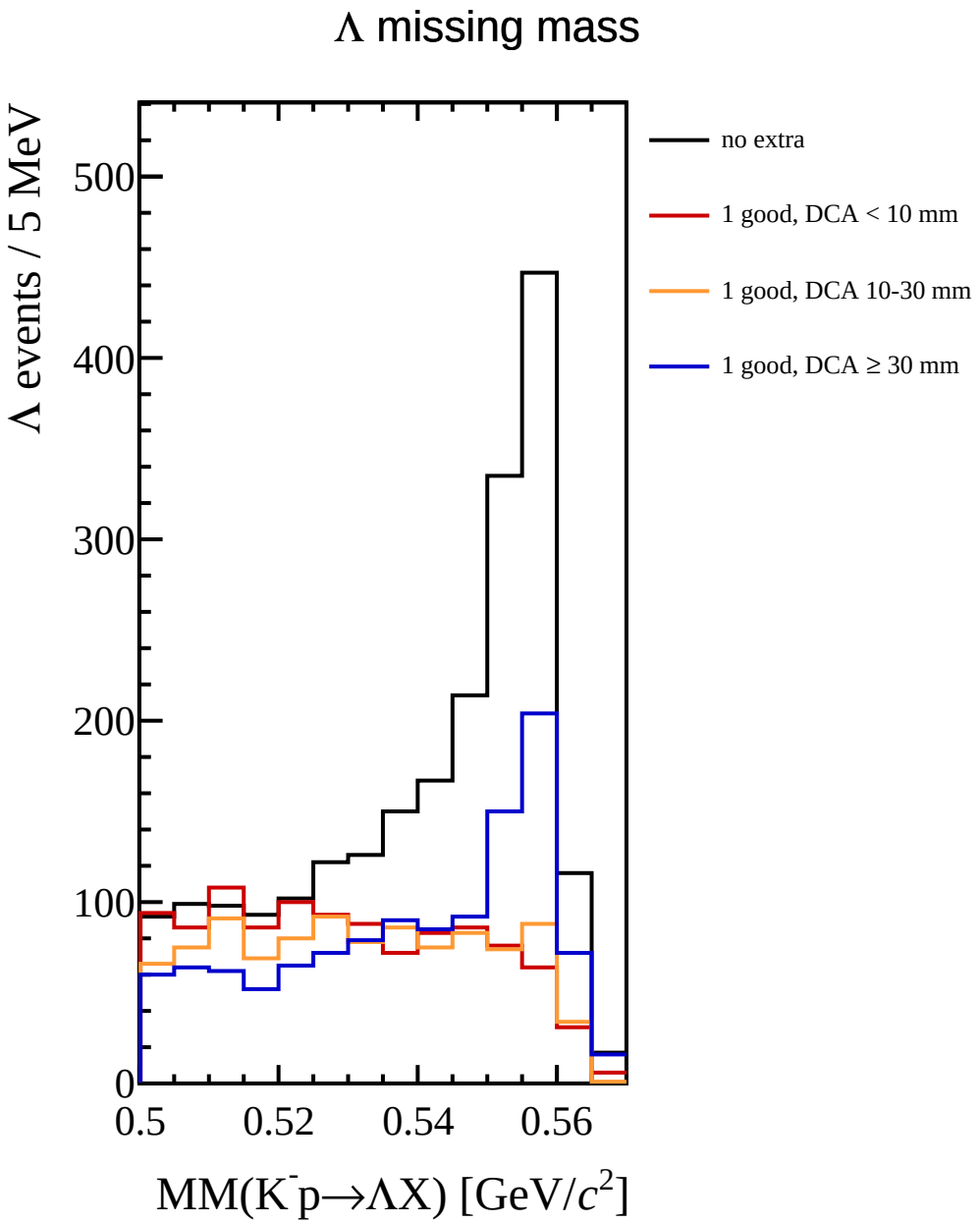


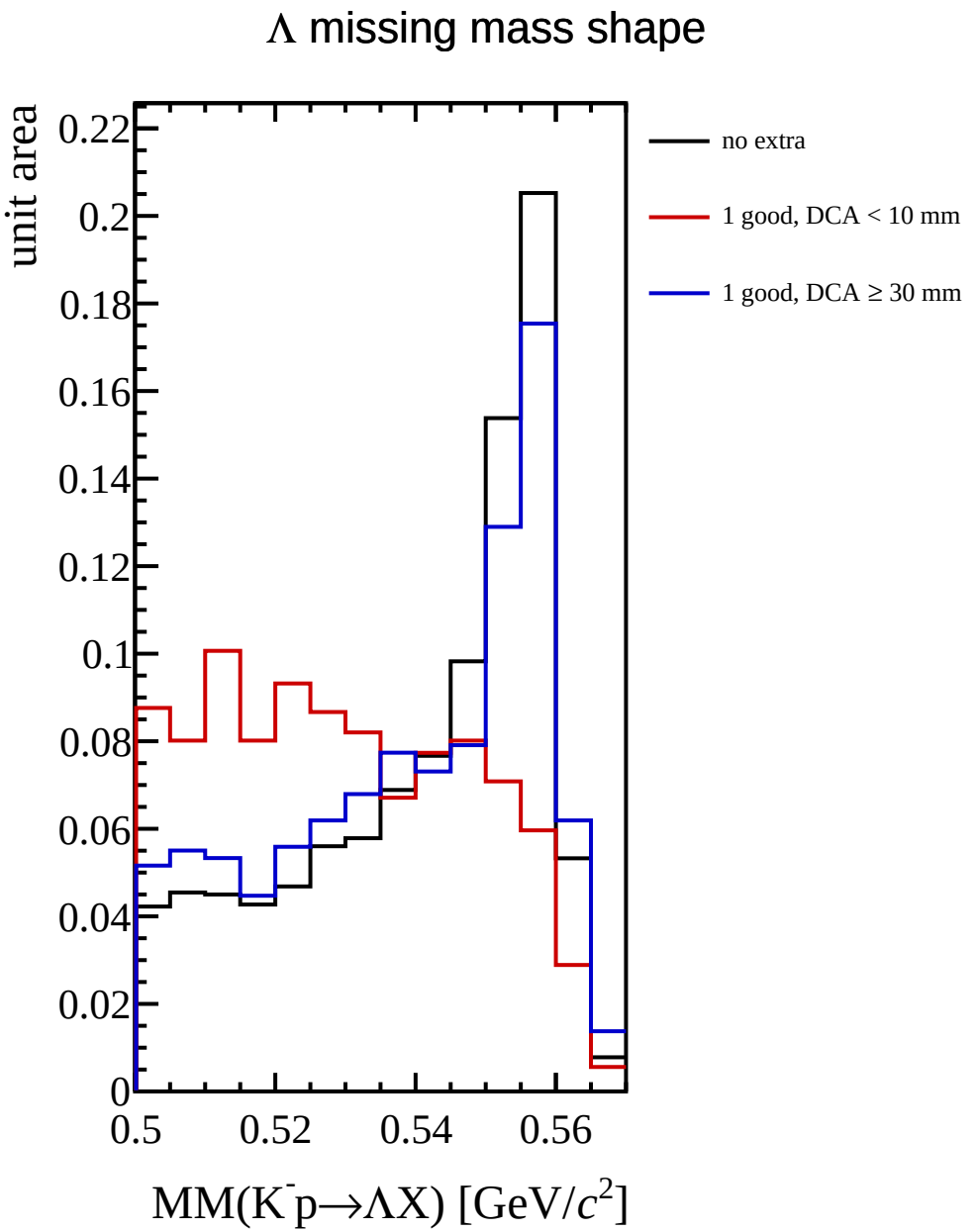

2 good, same charge: DCA to the production vertex
tracks / 1 mm
DCA to the production vertex [mm]
≥ 59 mm


3+ good: how many from the vertex
Λ events
good extra tracks from the vertex (DCA < 10 mm)


3+ good: species of the tracks
tracks
π⁻
π⁺
p
K±
e/other


all good extra tracks: DCA to the production vertex
tracks / 1 mm
DCA to the production vertex [mm]
≥ 59 mm


Λ missing mass
Λ events / 5 MeV
500
400
300
200
100
0
0.5
0.52
0.54
0.56
MM(K⁻p→ΛX) [GeV/c^2]
no extra
1 good, DCA < 10 mm
1 good, DCA 10-30 mm
1 good, DCA ≥ 30 mm


Λ missing mass shape
unit area
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0.5
0.52
0.54
0.56
MM(K⁻p→ΛX) [GeV/c^2]
no extra
1 good, DCA < 10 mm
1 good, DCA ≥ 30 mm